Reading Guide

On the sensitivity of nuclear clocks to new physics

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Reorganised and annotated for a nuclear-physics paper-reading seminar.

The paper in one paragraph

The $\approx 8\,\text{eV}$ transition in ^{229}Th is the basis of a future *nuclear clock*. Its tiny energy is believed to be an accident: a Coulomb (electromagnetic) energy shift of order 100 keV–1 MeV between the ground state and the isomer almost exactly cancels an opposite nuclear (strong-force) shift. If ultralight dark matter makes the fine-structure constant α_{em} (or the strong coupling α_s) oscillate in time, the two big pieces oscillate but no longer cancel, so the tiny 8 eV number wobbles by a hugely *enhanced* fraction, $K = |\Delta E_{\text{em}}|/\Delta E$. The question is: how big is K really, and could it accidentally be small (no enhancement, "worst case")? The authors answer with two nuclear models. A classical *geometric* (deformed liquid-drop) model says $K \sim 10^4$–10^5 is typical and $K \approx 0$ requires fine-tuning unknown shape parameters at the percent level. A new quantum *d-wave halo* model (^{228}Th core plus one loosely bound neutron), which correctly predicts the measured differences in charge radius and quadrupole moment between the two states, gives $\Delta E_{\text{em}} \sim 100\,\text{keV}$ and therefore $K \approx 10^4$ with no worst case at all.

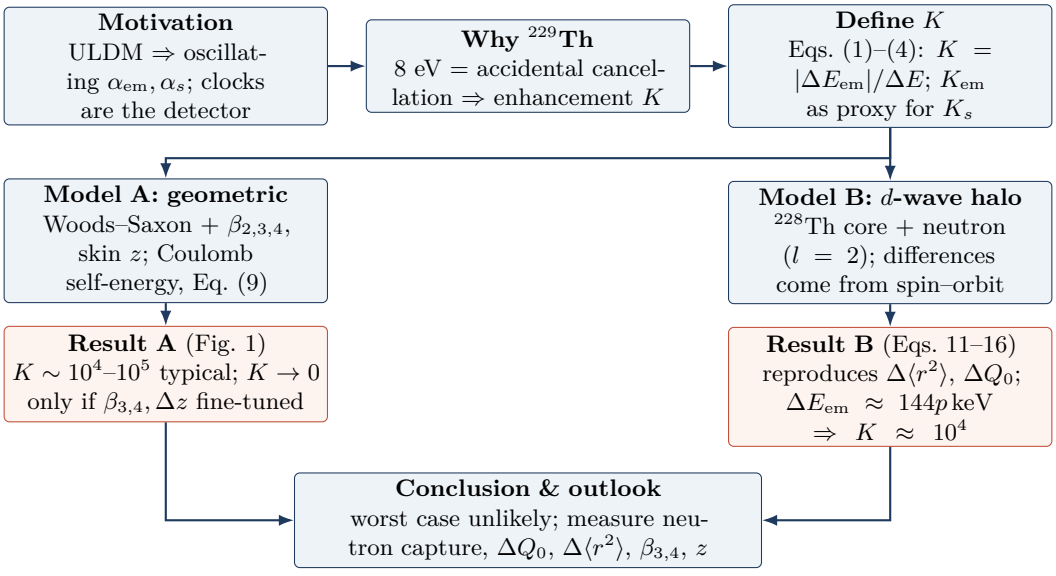
The paper in one equation

$$\underbrace{\Delta E}_{8.36\,\text{eV (measured)}} = \underbrace{\Delta E_{\text{em}}}_{\text{Coulomb, } \sim 10^2\text{–}10^3\,\text{keV}} + \underbrace{\Delta E_{\text{nuc}}}_{\text{strong, } \sim -\Delta E_{\text{em}}}$$

$$\Longrightarrow \quad \frac{\delta(\Delta E)}{\Delta E} = K\,\frac{\delta\alpha_{\text{em}}}{\alpha_{\text{em}}}, \qquad K \equiv \frac{|\Delta E_{\text{em}}|}{\Delta E}.$$

A relative change $\delta\alpha_{\text{em}}/\alpha_{\text{em}}$ in the fine-structure constant changes the clock frequency by K times as much. The whole paper is about pinning down K.

Roadmap of the argument



Paper section	Where it is in this guide
Introduction (p. 1)	Part 1 (motivation)
"The ^{229}Th Nucleus" (p. 2)	Part 2 (definitions of K, $K_{\rm em}$, K_s)
"The classical geometric model" (pp. 2–4) + Fig. 1	Part 3
"Th-229 as a Halo Nucleus" (pp. 4–5)	Part 4
"Summary and Outlook" (p. 5)	Part 5
(not in the paper)	Part 6: numbers, glossary, discussion questions

How to use this guide. The original is a 5-page *Letter*: dense, with the argument compressed into three short sections and one figure. This guide unpacks it into six parts that follow the paper's logic in the order you would want to *learn* it, not the order a journal requires. Each part opens with a box stating what that part of the paper is trying to achieve, followed by the physics, the equations (numbered as in the paper, e.g. "Eq. (3)"), and the result. Coloured boxes are: **Goal** (what the section is for), **Key idea** (the intuition), **Watch out** (caveats and places to be critical), and **HEP reader** (translations into experimental-particle-physics language). Part 6 collects the numbers, a glossary, and questions for the seminar discussion.

Part 1 Why anyone cares: dark matter, clocks, and a lucky nucleus

Goal of this part

Convince you that (i) a large class of dark-matter models predicts that the "constants" of nature oscillate slightly in time, (ii) comparing two clocks is the way to see it, and (iii) a ^{229}Th nuclear clock could be $\sim 10^4$–10^5 times more sensitive than any atomic clock *if* an assumed cancellation is real. The paper exists to test that "if".

1.1 Ultralight dark matter makes constants oscillate

Roughly 80% of the matter in the Universe is dark. One family of candidates is *ultralight dark matter* (ULDM): bosonic fields with masses far below an eV (axions, dilatons, relaxions, Higgs-portal scalars, Nelson–Barr models). Because the mass is so small, the galactic dark-matter density is not a gas of particles but a coherent classical field $\phi(t) \propto \cos(m_\phi t)$ oscillating at frequency m_ϕ. If ϕ couples to Standard Model fields, the effective values of couplings are modulated:

$$\alpha_{\rm em,s}(t) = \alpha_{\rm em,s}\big[1 + F(t)\, g_{\rm em,s}\big],$$

where g is the (unknown, tiny) DM coupling and $F(t)$ carries the oscillation. Scalars couple linearly to hadron masses; axions couple quadratically. Most of the theoretically appealing models couple predominantly to the *QCD sector* (so nuclear parameters oscillate) and only subdominantly to QED (so $\alpha_{\rm em}$ oscillates less).

1.2 Clock comparison as a detector

A single clock cannot see a change in $\alpha_{\rm em}$, because every frequency reference shifts together. Two clocks whose transition frequencies depend *differently* on $\alpha_{\rm em}$ or on nuclear parameters will drift relative to each other, and the ratio ν_1/ν_2 oscillates at the DM frequency. Current limits come from atomic and molecular spectroscopy, optical cavities, and mechanical oscillators. Optical atomic clocks are extremely precise but almost blind to *nuclear* parameters: the nucleus enters only through hyperfine structure ($\sim 10^{-6}$), reduced mass ($\sim 10^{-5}$) and the charge radius ($\sim 10^{-3}$).

For an experimental HEP reader

Think of a clock-ratio measurement exactly like a ratio of cross sections or a double ratio in a heavy-ion analysis: common systematics (here, anything that shifts *both* clocks) cancel, and you are looking for a small time-dependent residual. The "sensitivity coefficient" K introduced below plays the role of a signal amplification factor: for a fixed instrumental precision on ν_1/ν_2, your reach in the coupling g scales as $1/K$.

1.3 Why ^{229}Th: an 8 eV nuclear transition

Nuclear excitation energies are usually tens of keV to MeV. ^{229}Th is the outlier: its first excited state (the isomer ^{229m}Th) lies at

$$\Delta E = 8.355\,733\,554\,021(8)\,\text{eV},$$

i.e. in the vacuum ultraviolet, where lasers exist. It was laser-excited for the first time in 2024 and its frequency has been measured relative to a ^{87}Sr atomic clock. A clock built on this line would be the first *nuclear* clock and would couple directly to nuclear parameters with $O(1)$ strength, instead of the 10^{-3}–10^{-6} suppression of atomic clocks.

1.4 The accidental cancellation and the enhancement factor

Why is the transition so low? The accepted explanation is that the energy difference splits into an electromagnetic piece and a strong-interaction piece,

$$\Delta E = \Delta E_{\text{em}} + \Delta E_{\text{nuc}}, \qquad |\Delta E_{\text{em}}| \sim |\Delta E_{\text{nuc}}| \gg \Delta E,$$

which nearly cancel. Individually, each is a normal nuclear number, of order 100 keV to 1 MeV. If new physics changes α_{em} by a relative amount $\delta\alpha_{\text{em}}/\alpha_{\text{em}}$, only the electromagnetic piece responds, $\delta(\Delta E_{\text{em}}) = \Delta E_{\text{em}}\,\delta\alpha_{\text{em}}/\alpha_{\text{em}}$, but this shift is compared to the tiny total ΔE. Hence the relative frequency shift of the clock is enhanced by

$$K = \frac{|\Delta E_{\text{em}}|}{\Delta E} \sim \frac{10^5\text{–}10^6\,\text{eV}}{8\,\text{eV}} \sim 10^4\text{–}10^5.$$

Figure 1 illustrates this.

Watch out

The cancellation is an *assumption*, not an observation. ΔE_{em} is not measurable on its own, and no nuclear model can compute an energy difference to 8 eV precision. Logically, it could be that ΔE_{em} and ΔE_{nuc} are *each* of order eV, in which case $K \sim O(1)$ and the nuclear clock has no advantage. The authors call this the *worst case scenario*. The two questions the paper sets out to answer are:

Q1. What is the expected enhancement factor K?

Q2. How likely is the worst case, where ΔE_{em} is accidentally close to zero?

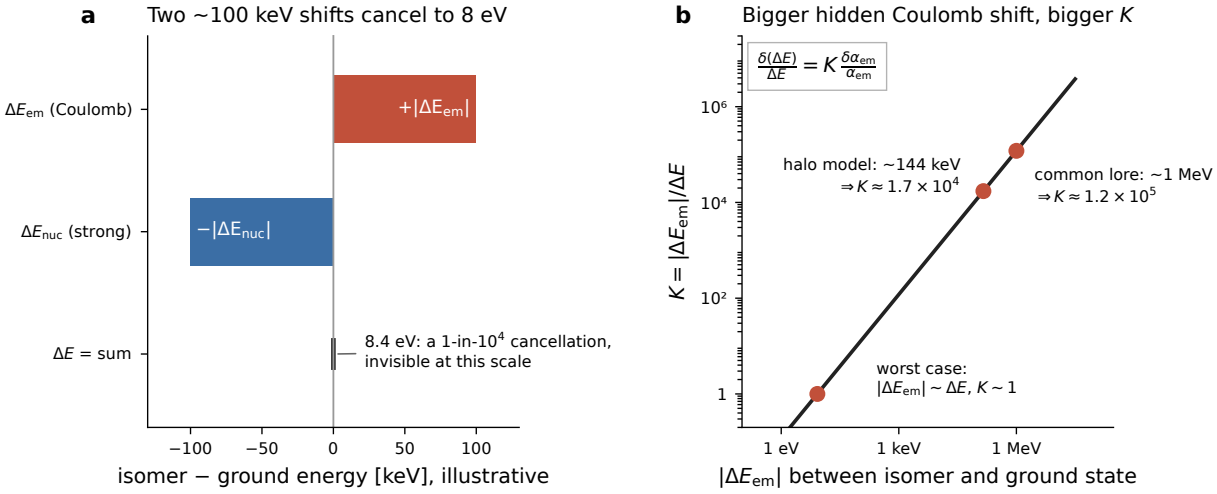


Figure 1: **(a)** The cancellation picture: two energy shifts of order 100 keV (illustrative values) between isomer and ground state almost exactly cancel, leaving the 8.4 eV transition. **(b)** The enhancement factor is simply the ratio of the hidden Coulomb shift to the observed transition energy. The three points mark the "worst case" (no cancellation, $K \sim 1$), the halo-model result of this paper (~ 144 keV, $K \approx 1.7 \times 10^4$), and the older estimate (~ 1 MeV, $K \approx 1.2 \times 10^5$).

Part 2 Setting up the problem: what exactly is being computed

Goal of this part

Translate "sensitivity to new physics" into a single number that a nuclear model can produce: the electromagnetic enhancement factor $K_{\rm em}$ (Eq. 3). Argue that it is also a good proxy for the sensitivity to changes in the *strong* coupling, K_s, which is what most DM models actually affect but which nobody can compute.

2.1 From coupling variations to a frequency shift (Eqs. 1–3)

Start from Eq. (1), $\Delta E = \Delta E_{\rm em} + \Delta E_{\rm nuc}$. If new physics shifts $\alpha_{\rm em}$ and α_s, then (Eq. 2)

$$\delta(\Delta E) = \frac{\partial \Delta E_{\rm em}}{\partial \alpha_{\rm em}}\,\delta\alpha_{\rm em} + \frac{\partial(\Delta E_{\rm em} + \Delta E_{\rm nuc})}{\partial \alpha_s}\,\delta\alpha_s.$$

Two things to notice: only $\Delta E_{\rm em}$ depends on $\alpha_{\rm em}$ (the strong part does not know about photons at this level), while *both* pieces depend on α_s (the strong coupling sets the nuclear size and density, which the Coulomb energy also depends on).

Focusing on the electromagnetic term and using the fact that a Coulomb energy is linear in $\alpha_{\rm em}$ (it is $\propto e^2$), the paper defines (Eq. 3)

$$K_{\rm em} \equiv \frac{1}{\Delta E}\frac{\partial \Delta E_{\rm em}}{\partial \log \alpha_{\rm em}} \simeq \frac{\Delta E_{\rm em}}{\Delta E}.$$

The logarithmic derivative is just "relative change in ΔE per relative change in $\alpha_{\rm em}$". The second equality is the crucial simplification: **to get K you only need $\Delta E_{\rm em}$ itself**, the Coulomb-energy difference between the two nuclear states. That is what both models in the paper compute. The enhancement factor used everywhere else is $K \equiv |K_{\rm em}|$.

Why Coulomb energy is "linear in $\alpha_{\rm em}$"

The electrostatic self-energy of a charge distribution is $E_C = \frac{1}{2}\iint \rho\rho'/|\mathbf{r}-\mathbf{r}'|$ with $\rho \propto e$, so $E_C \propto e^2 \propto \alpha_{\rm em}$. If $\alpha_{\rm em} \to \alpha_{\rm em}(1+\epsilon)$ then $\Delta E_{\rm em} \to \Delta E_{\rm em}(1+\epsilon)$ and $\delta(\Delta E) = \epsilon\,\Delta E_{\rm em}$. Dividing by ΔE gives ϵK. This holds only if the nuclear *shape* does not itself change with $\alpha_{\rm em}$; that back-reaction is second order and neglected.

2.2 The strong-sector proxy argument (Eq. 4)

Most attractive ULDM candidates couple to QCD, so the physically relevant quantity is

$$K_s = K_s^{\rm em} + K_s^{\rm nuc}, \qquad K_s^{\rm em} \equiv \frac{1}{\Delta E}\frac{\partial \Delta E_{\rm em}}{\partial \log \alpha_s}, \quad K_s^{\rm nuc} \equiv \frac{1}{\Delta E}\frac{\partial \Delta E_{\rm nuc}}{\partial \log \alpha_s}.$$

Nobody can compute how $\Delta E_{\rm nuc}$ responds to α_s. The paper's argument (Eq. 4) is: the Coulomb energy depends on α_s only through nuclear geometry (radius, density), which scales as some power of the QCD scale, so

$$K_s^{\rm em} \sim \beta\,\frac{\Delta E_{\rm em}}{\Delta E} \sim \beta\,K_{\rm em}, \qquad \beta = O(1).$$

Then, *unless* $K_s^{\rm em}$ and $K_s^{\rm nuc}$ happen to cancel each other, $K_s \sim K_{\rm em}$ up to an order-one factor. Cancellation would require $\Delta E_{\rm em}$ and $\Delta E_{\rm nuc}$ to have *exactly* the same functional dependence on α_s, which EFT treatments of nuclear forces do not predict (they scale differently with the QCD scale). This is why the paper claims its conclusions apply to DM coupled to either sector.

Watch out

This proxy argument is a plausibility argument, not a calculation. The authors flag it themselves in the outlook ("verifying this within concrete frameworks is key"). It is a good seminar discussion point: the entire relevance to *QCD-coupled* dark matter rests on $\beta = O(1)$ and no accidental cancellation in K_s.

2.3 Two things the paper neglects on purpose

- **Electrons.** The atomic electron cloud penetrates the nucleus and shifts the transition, but the effect is a *field shift* $F\,\delta\langle r^2\rangle \approx 10^{-7}\,\text{eV}$ (with $F \sim \text{GHz/fm}^2$), eight orders of magnitude below ΔE. Dropped.
- **Exact nuclear structure.** The paper states plainly that computing ΔE_{em} to eV accuracy is "considered absurd in nuclear physics". The aim is a *preferred range* for ΔE_{em} and a probability estimate for the worst case, not a precise number.

Part 3 Model A: the classical geometric nucleus

Goal of this part

Take the simplest picture of a nucleus (a uniformly charged, slightly deformed drop with a fuzzy surface), fix what can be fixed from data (charge radius, quadrupole moment), scan over what cannot (higher deformations β_3, β_4 and the skin-thickness difference Δz), compute the Coulomb self-energy difference between isomer and ground state, and ask how much of parameter space gives $K \approx 0$.

3.1 The picture: a deformed charged drop with a soft edge

The charge density is a Woods–Saxon (Fermi-function) profile, Eq. (5):

$$\rho(r,\theta) = \rho_0 \left[1 + \exp\left(\frac{r - R(\theta)}{z}\right)\right]^{-1},$$

where $z \approx 0.5\,\text{fm}$ is the *skin thickness* (how quickly the density falls from its central value to zero), $R(\theta)$ is the direction-dependent radius, and ρ_0 is set by requiring the total charge to be Ze with $Z = 90$. Deformation enters through Eq. (6):

$$\frac{R(\theta)}{R_0} = 1 + \beta_2 Y_{20}(\theta) + \beta_3 Y_{30}(\theta) + \beta_4 Y_{40}(\theta) + \ldots,$$

with β_2 = quadrupole (football shape), β_3 = octupole (pear shape), β_4 = hexadecapole (squarer or more diamond-like shape). Azimuthal symmetry is assumed, so only $m = 0$ harmonics appear. Figure 2 shows what these knobs do.

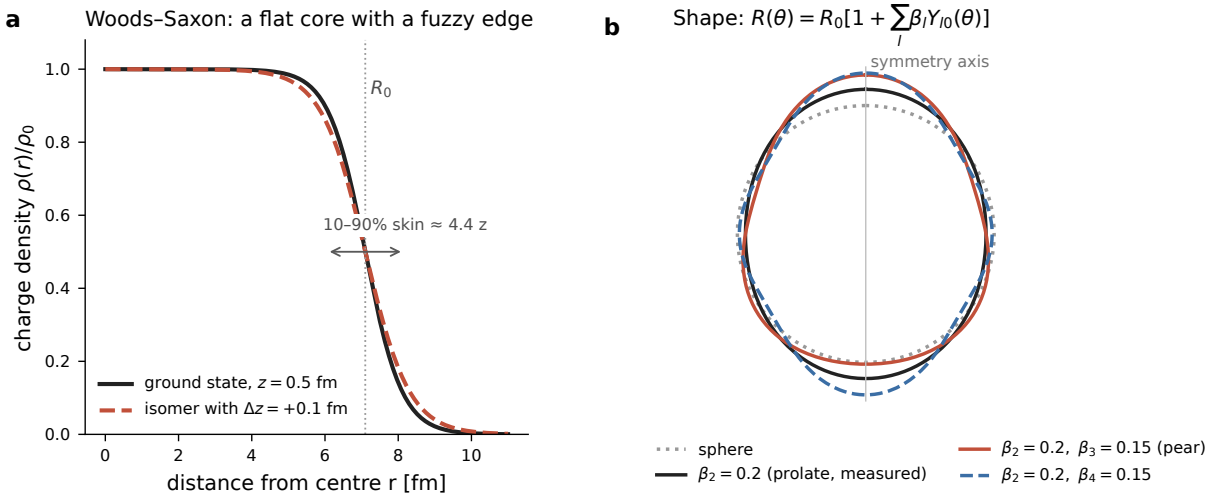


Figure 2: **(a)** Woods–Saxon radial profile: flat inside, falling to zero over a distance $\sim 4.4z$ around R_0. A different skin thickness in the isomer (Δz) redistributes charge at the surface and changes the Coulomb energy. **(b)** The angular shape for the measured quadrupole deformation $\beta_2 \approx 0.2$ and for illustrative octupole (β_3) and hexadecapole (β_4) admixtures. These higher moments are unmeasured for ^{229}Th and are the main unknowns of Model A.

3.2 What is measured and fixes the model (Eqs. 7–8)

Two moments of the charge density are experimentally accessible for both states:

$$\text{mean-square charge radius:} \quad \langle r^2 \rangle = \frac{1}{eZ} \int d^3r\, r^2 \rho(r,\theta), \qquad \text{Eq. (7)}$$

$$\text{intrinsic quadrupole moment:} \quad Q_0 = \frac{1}{eZ} \int d^3r\, r^2 \rho(r,\theta)\, [3\cos^2\theta - 1], \qquad \text{Eq. (8)}$$

with measured values $\sqrt{\langle r^2 \rangle} = 5.7557(143)\,\text{fm}$ and $Q_0 = 9.8(1)\,\text{fm}^2$. The radius fixes mainly R_0; the quadrupole moment fixes $\beta_2 \approx 0.2$. Crucially, the *differences* between isomer and ground state are also measured: $\Delta\langle r^2 \rangle = 0.012(2)\,\text{fm}^2$ and (from the 2024 frequency-ratio paper) $\Delta Q_0/Q_0 = 0.0179$, i.e. $Q_0^{\text{iso}} - Q_0^{\text{gs}} = 0.173(9)\,\text{fm}^2$.

Intrinsic vs. spectroscopic quadrupole moment

Experiments usually report the *spectroscopic* moment $Q_s = Q_0 \dfrac{j(2j-1)}{(j+1)(2j+3)}$, which is what you measure in the lab frame for a state of spin j. The *intrinsic* Q_0 is the moment in the body-fixed frame of the deformed nucleus. Since the ground state ($j = 5/2$) and isomer ($j = 3/2$) have different j, the conversion factors differ; the paper works with Q_0 throughout.

3.3 What is *not* measured: the free parameters

Three quantities affect ΔE_{em} but are unconstrained:

- $\beta_3^{\text{gs}}, \beta_3^{\text{iso}}$ (octupole) and $\beta_4^{\text{gs}}, \beta_4^{\text{iso}}$ (hexadecapole), scanned over $[0, 0.3]$, i.e. up to the size of β_2;
- Δz, the skin-thickness difference between the two states, scanned over roughly $\pm 0.2\,\text{fm}$;
- $\Delta Q_0/Q_0$ itself is varied across a range to show how sensitively K depends on it.

Two improvements over earlier work (Fadeev et al. 2020; Berengut et al. 2009) are made explicit: the authors do *not* assume equal volume for the two states (they let R_0 float and solve numerically rather than expanding in small β), and they include β_4 and Δz *simultaneously* with β_3.

3.4 The computation (Eq. 9)

The electromagnetic energy of each state is approximated by its direct Coulomb self-energy,

$$E_{\text{em}} \simeq E_C \equiv \frac{1}{2} \int d^3r\, d^3r'\, \frac{\rho(r,\theta)\, \rho(r',\theta')}{|\mathbf{r} - \mathbf{r}'|},$$

evaluated numerically for the ground-state and isomer parameter sets; then $\Delta E_{\text{em}} = E_C^{\text{iso}} - E_C^{\text{gs}}$ and $K = |\Delta E_{\text{em}}|/\Delta E$.

For an experimental HEP reader

This is a six-dimensional integral of a classical charge distribution: no quantum mechanics, no Pauli exchange, no nucleon–nucleon correlations. It is the nuclear analogue of computing an electrostatic energy of a macroscopic charged object. The authors are open that this is a caricature; its virtue is that it is transparent and that it was the model used in the literature to quote $K \sim 10^5$, so it is the natural place to test how fragile that number is.

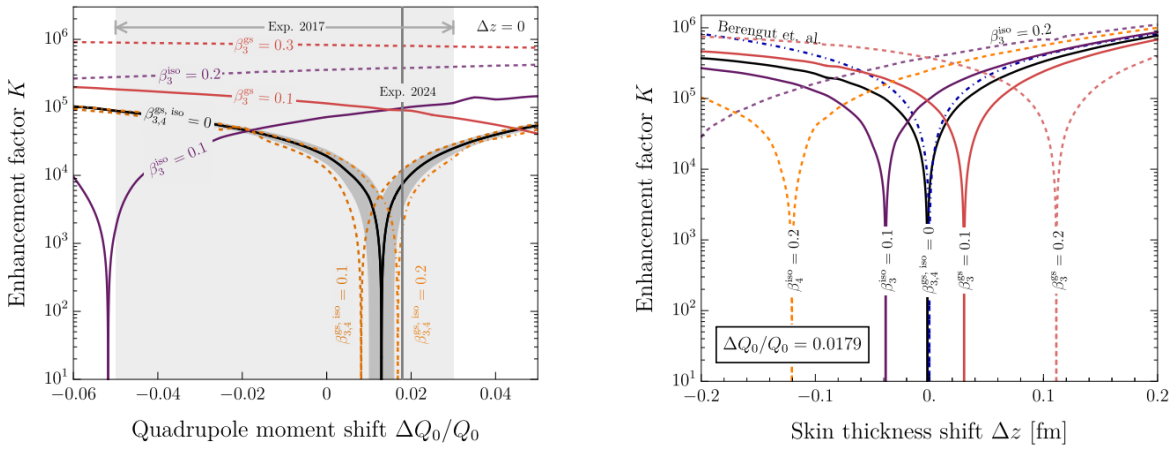


Figure 3: Figure 1 of Caputo et al. (reproduced from arXiv:2407.17526 for study purposes). **Left:** K versus the relative quadrupole-moment difference $\Delta Q_0/Q_0$ for various choices of the unmeasured β_3, β_4 (at $\Delta z = 0$). Vertical grey line: the 2024 measurement; light band: 1σ of the older 2017 measurement; dark band: 2σ from $\langle r^2 \rangle$ for $\beta_{3,4} = 0$. **Right:** K versus the skin-thickness difference Δz at the measured $\Delta Q_0/Q_0 = 0.0179$. The blue dot-dashed line is the approximate formula of Berengut et al. (2009).

3.5 Results: Fig. 1 of the paper, walked through

Left panel (quadrupole shift). Each curve is K as a function of $\Delta Q_0/Q_0$ for a given choice of the unmeasured $\beta_{3,4}$. Every curve has a sharp *funnel* where $K \to 0$: that is the value of $\Delta Q_0/Q_0$ at which the Coulomb energy change from the shape difference happens to vanish. Read it as follows:

- For $\beta_{3,4} = 0$ (black curve) the funnel sits at $\Delta Q_0/Q_0 \approx 0.013$, uncomfortably close to the measured 0.018 (grey line); but even then, at the measured value $K \sim 10^4$.
- Turning on any β_3 or β_4 moves the funnel. With $\beta_3^{\rm gs} = 0.3$, $K \sim 10^6$ for all $\Delta Q_0/Q_0$; with $\beta_3^{\rm iso} = 0.1$ the funnel moves far to the left (-0.05).
- Getting $K \approx 0$ therefore requires the *combination* of β_3, β_4 and $\Delta Q_0/Q_0$ to conspire so that a funnel lands exactly on the measured line. Since $\Delta Q_0/Q_0$ is now known to $\sim 5\%$, this is a percent- to per-mille-level coincidence. Away from funnels, $K \sim 10^4$–10^5.

Right panel (skin thickness). Same story with Δz as the variable at fixed $\Delta Q_0/Q_0 = 0.0179$. Each $\beta_{3,4}$ choice produces a funnel at some Δz between -0.12 and $+0.11$ fm; $\Delta z = 0$ with $\beta_{3,4} = 0$ sits essentially on a funnel, which is why the earlier simple estimate (Berengut et al.) and this one agree only near $\Delta z \approx 0$. For generic Δz and $\beta_{3,4}$, $K \sim 10^4$–10^5.

Answer from Model A

Q1: typical $K \sim 10^4$–10^5. **Q2:** the worst case $K \approx 0$ is possible but requires the unknown shape parameters to be tuned at the percent or per-mille level so that a zero of $\Delta E_{\rm em}$ coincides with the measured quadrupole shift. The authors judge this "fairly unlikely". The weakness is that the model has several unmeasured parameters, so "unlikely" cannot be turned into a probability.

Known shortcomings of Model A

Classical; no exchange (Fock) term; no quantum shell structure; the electromagnetic spin–orbit contribution, which the paper's own Model B says dominates the *difference* between the states,

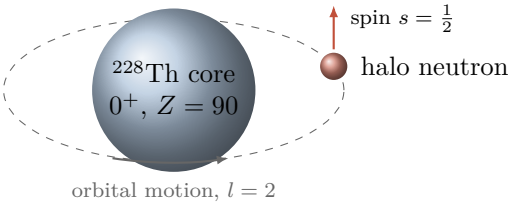
is absent. Model A is kept because it is what the literature used and because it is the only framework in which $\beta_{3,4}$ and z can be discussed at all.

Part 4 Model B: ^{229}Th as a d-wave halo nucleus

Goal of this part

Replace the charged drop by a quantum model tailored to the *difference* between the two states: ^{229}Th = spin-0 ^{228}Th core + one weakly bound neutron in an $l = 2$ orbit. Ground state and isomer are then the two spin–orbit partners $j = 5/2$ and $j = 3/2$ of the same orbit. Validate the model against the measured $\Delta\langle r^2\rangle$ and ΔQ_0, then compute the spin–orbit contribution to ΔE_{em} and hence K.

4.1 The picture



$\boldsymbol{j = l + s} = \frac{5}{2}^+$ ground state

$\boldsymbol{j = l - s} = \frac{3}{2}^+$ isomer ^{229m}Th

Same orbit, same radial wave function $u(r)$; the two states differ only in how the neutron spin is aligned with its orbit.

Why is this plausible? (i) The neutron separation energy of ^{229}Th ($S_n \approx 5.2\,\text{MeV}$) is noticeably smaller than that of ^{228}Th ($\approx 7.1\,\text{MeV}$), the signature of one loosely attached neutron; (ii) charge-radius systematics along the Th chain are consistent with a core-plus-neutron picture; (iii) the observed spins and parities $5/2^+$ and $3/2^+$ are exactly an $l = 2$ spin–orbit doublet.

"Halo EFT" here is a model, not a controlled EFT

A proper halo effective field theory needs a separation of scales between the halo neutron's binding and the core's excitation. Here the expansion parameter is estimated as $\sqrt{5.2/7.1} \approx 0.86$: far from small. Moreover, for d-waves the centrifugal barrier makes observables depend strongly on short-distance physics (the neutron wave function inside the core), which shows up as counter-terms of unknown size. The authors state this explicitly and use the framework as a *model* whose structure is borrowed from EFT.

4.2 The key structural insight

At leading order in halo EFT, the $j = 3/2$ and $j = 5/2$ partners are *degenerate*, and therefore all their differences (charge radius, quadrupole moment, Coulomb energy, …) *vanish* at leading order. The transition energy is a difference, so the model naturally isolates the physics of the isomeric transition: everything is driven by the one thing that distinguishes the states, the *spin–orbit* interaction between the halo neutron's spin and its $l = 2$ orbital motion. This is why the model can be predictive for differences even though it cannot predict absolute energies.

4.3 How a neutral particle changes the charge density (Eq. 10)

A neutron carries no net charge, yet it has an anomalous magnetic moment $\mu_n = -1.91\,\mu_N$. A moving magnetic dipole has, in the nucleus frame, an effective electric charge distribution (the relativistic "spin–orbit" or Darwin–Foldy-type term). Its charge density is

$$\rho_{\text{SO}} = \frac{\mu_n}{2m_n^2}\, i\,\sigma_{\alpha\beta} \cdot \left[(\nabla\Phi_\alpha^\dagger) \times (\nabla\Phi_\beta)\right],$$

with Φ the two-component neutron spinor. Because it involves $\boldsymbol{\sigma}$ and the orbital gradient, it is proportional to $\boldsymbol{l} \cdot \boldsymbol{s}$ and therefore has *opposite sign* for $j = l + \frac{1}{2}$ and $j = l - \frac{1}{2}$. Inserting ρ_{SO} into the same moment formulas as before (Eqs. 7–8) gives the model's prediction for the *differences* between the states.

4.4 Validation against data (Eqs. 11–13)

Observable difference (isomer − ground)	Halo-model prediction	Measured	Ratio
$\Delta\langle r^2\rangle$ [fm^2]	0.0047 (Eq. 11)	0.012(2)	≈ 0.4
ΔQ_0 [fm^2]	0.185 (Eq. 12)	0.173(9) (Eq. 13, 2024)	1.07

The quadrupole-moment difference is reproduced to within 7%, with *no free parameter*. The radius difference is a factor ~ 2 low, which the authors consider excellent given the model's crudeness; they note it could reflect an in-medium renormalisation of the neutron's magnetic moment. Importantly, both of these quantities are *insensitive to the ultraviolet*, i.e. to the poorly known neutron wave function inside the core, because they are weighted by r^2 and dominated by the halo tail.

For an experimental HEP reader

This is the paper's "closure test". The model was not fitted to ΔQ_0; it predicted it, and the 2024 measurement landed on top of it. In an analysis you would call this a successful validation of the model in a control region before using it in the signal region ($\Delta E_{\rm em}$), with the important caveat that the signal-region quantity is *more* UV-sensitive than the control-region quantities.

4.5 The energy: spin–orbit splitting in the core's Coulomb field (Eqs. 14–16)

The Coulomb energy difference between the two states is the electromagnetic spin–orbit energy of the halo neutron moving in the Coulomb potential $V_C(r)$ of the ^{228}Th core:

$$E_{\rm SO} = \frac{e\,\mu_n}{2m_n^2}\Big[j(j+1) - l(l+1) - \tfrac{3}{4}\Big]\int_0^\infty \frac{[u(r)]^2}{r}\,\frac{dV_C}{dr}\,dr.$$

The bracket is just $2\langle \boldsymbol{l}\cdot\boldsymbol{s}\rangle$: it equals $+2$ for $j = 5/2$ and -3 for $j = 3/2$, so the two states are split by $5\times$(the integral). The radial wave function $u(r)$ is the bound-state d-wave tail, Eq. (15),

$$u(r) = A(r)\,e^{-\gamma r}\Big(1 + \frac{3}{\gamma r} + \frac{3}{(\gamma r)^2}\Big), \qquad \gamma = \sqrt{2m_n E_B}, \quad E_B = 5.2\,\text{MeV},$$

where $A(r)$ tends at large r to the *asymptotic normalisation constant* (ANC) A, and its short-distance behaviour is unknown. The same $u(r)$ is used for both states (their separation energies are nearly equal and the spin–orbit force is a small perturbation).

Why the answer is not hopelessly UV-sensitive

The halo wave function lives at $r \sim 1/\gamma \sim$ a few fm. Inside a Woods–Saxon-like core the charge density is roughly *constant*, so the Coulomb potential is harmonic-oscillator-like and $dV_C/dr \propto r$. The $1/r$ in the integrand is therefore cancelled, and the integral is dominated by the well-understood tail rather than by the origin. The authors regulate the residual short-distance behaviour with a cut-off l and find the result almost independent of it for $l \lesssim 4$ fm.

The result (Eq. 16) is

$$\Delta E_{\rm SO} = E_{\rm SO}\big|_{j=5/2} - E_{\rm SO}\big|_{j=3/2} \approx 144\,p\,\text{keV},$$

where $p = O(1)$ depends on the ANC and encodes the probability of finding the halo neutron inside the core; a full nuclear-structure calculation would fix it. Hence

$$\Delta E_{\rm em} \sim 100\,\text{keV} \quad\Longrightarrow\quad K \approx \frac{10^5\,\text{eV}}{8\,\text{eV}} \approx 10^4.$$

Answer from Model B

Q1: $K \approx 10^4$ (i.e. $\gtrsim 10^4$, since p could be somewhat larger than 1). **Q2:** the worst case *never* occurs: the spin–orbit splitting has a definite sign and a definite magnitude of order 100 keV; there is no free parameter that can drive it to zero. Model B also predicts the two measurable differences correctly, whereas Model A needs them as input.

Part 5 Conclusions and what the authors want measured next

Goal of this part

State the bottom line and turn the model uncertainties into concrete experimental requests, so that the nuclear-clock community can sharpen K.

5.1 Bottom line

- In the classical geometric model, $K \to 0$ is possible but requires per-cent- to per-mille-level fine-tuning of unmeasured shape parameters; typical values are 10^4–10^5.
- In the more realistic d-wave halo model, validated on $\Delta\langle r^2\rangle$ and ΔQ_0, $K \gtrsim 10^4$ always.
- Via the proxy argument of Part 2, the same conclusions are claimed for DM coupled to the strong sector, barring an exact coincidence in the α_s-dependence of $\Delta E_{\rm em}$ and $\Delta E_{\rm nuc}$.

5.2 Requested measurements and theory follow-ups

Proposal	**Why it matters for K**
Neutron-capture cross sections on ^{228}Th into the ground state and into the isomer	The capture amplitude into a d-wave halo is $\propto$ the ANC A, which controls p and hence the size of $\Delta E_{\rm SO}$. Halo EFT predicts the ratio $\sigma_{\rm cap}^{\rm iso} = \frac{2}{3}\,\sigma_{\rm cap}^{\rm gs}$; measuring it tests the assumption that the two states share one $u(r)$.
Confirm the 2024 value of $\Delta Q_0/Q_0$	The halo model's cleanest prediction; also the variable along which the geometric model's funnels sit.
Improve $\Delta\langle r^2\rangle$	Currently the one place where the halo model is off by a factor ~ 2.
Devise probes of β_3, β_4 and the skin thickness z	They determine K in the geometric model; presently unconstrained.
Theory: check $K_s \sim K_{\rm em}$ in concrete nuclear EFTs	The strong-sector relevance rests on $\Delta E_{\rm em}$ and $\Delta E_{\rm nuc}$ scaling differently with the QCD scale.
Theory: a microscopic model for $A(r)$	Would remove the regulator and fix p.

Part 6 Seminar toolkit: numbers, glossary, and questions

6.1 Key numbers

Quantity	Value	Source / role
Transition energy ΔE	$8.355\,733\,554\,021(8)$ eV	Zhang et al. 2024; the "small number"
Spins/parities	gs $5/2^+$, isomer $3/2^+$	$l = 2$ spin–orbit doublet
Charge radius $\sqrt{\langle r^2 \rangle}$	$5.7557(143)$ fm	fixes R_0 in Model A
Intrinsic quadrupole Q_0	$9.8(1)\,\mathrm{fm}^2$	fixes $\beta_2 \approx 0.2$
$\Delta\langle r^2 \rangle$ (iso − gs)	$0.012(2)\,\mathrm{fm}^2$	Model B predicts 0.0047
ΔQ_0 (iso − gs)	$0.173(9)\,\mathrm{fm}^2$ $(\Delta Q_0/Q_0 = 0.0179)$	Model B predicts 0.185
Skin thickness z	~ 0.5 fm	Model A; Δz scanned ± 0.2 fm
Scan range for $\beta_{3,4}$	$[0, 0.3]$	Model A unknowns
Neutron separation energies	^{229}Th: 5.2 MeV; ^{228}Th: 7.1 MeV	halo picture; E_B in Eq. (15)
Halo expansion parameter	$\sqrt{5.2/7.1} \approx 0.86$	why Model B is "a model"
Neutron magnetic moment	$\mu_n = -1.91\,\mu_N$	source of ρ_{SO}
Spin–orbit splitting	$\Delta E_{\mathrm{SO}} \approx 144\,p$ keV	Eq. (16), $p = O(1)$
Electron field shift	$\approx 10^{-7}$ eV	neglected
Enhancement factor K	10^4–10^5 (A); $\gtrsim 10^4$ (B)	the paper's result

6.2 Glossary

Isomer
A long-lived excited nuclear state. ^{229m}Th is long-lived because the transition is a low-energy M1 and radiatively suppressed; this is what makes it a clock candidate.

Nuclear clock
A frequency standard locked to a nuclear rather than an atomic-electron transition. Advantage: the nucleus is small and shielded from external fields, and it is directly sensitive to nuclear (QCD) parameters.

Ultralight dark matter (ULDM)
Bosonic DM with mass $\ll 1$ eV behaving as a coherent oscillating classical field; induces time-oscillation of effective coupling constants.

Enhancement / sensitivity factor K
$|\Delta E_{\mathrm{em}}|/\Delta E$: how much more a relative change in α_{em} shifts the clock frequency compared to a generic transition.

Fine-tuning / cancellation
Two large terms of opposite sign summing to a much smaller result. Here $\Delta E_{\mathrm{em}} \approx -\Delta E_{\mathrm{nuc}}$ to one part in 10^4.

Woods–Saxon density
$\rho_0/[1 + e^{(r-R)/z}]$: constant interior, surface falling over a distance set by the skin thickness z.

$\beta_2, \beta_3, \beta_4$
Quadrupole (spheroidal), octupole (pear), hexadecapole deformation coefficients in $R(\theta) = R_0[1 + \sum \beta_l Y_{l0}]$.

Intrinsic vs. spectroscopic quadrupole moment
Q_0 in the body frame vs. Q_s in the lab frame; related by a spin-dependent factor.

Halo nucleus
A core plus one or two weakly bound nucleons whose wave function extends far outside the core (e.g. ^{11}Li, ^{11}Be, ^{17}C). ^{229}Th is proposed as a heavy, d-wave example.

Halo EFT
Effective field theory exploiting the separation between the halo nucleon's small binding and the core's higher excitation energies; expansion in their ratio.

Asymptotic normalisation constant (ANC)
The coefficient A of the exponential tail of a bound-state wave function; sets capture and breakup cross sections; here it controls p.

Spin–orbit charge density ρ_{SO}
The effective charge density of a moving neutral particle with a magnetic moment; $\propto \boldsymbol{l} \cdot \boldsymbol{s}$, hence opposite for the two j partners.

UV sensitivity
Dependence of a low-energy prediction on unknown short-distance physics; here, on the neutron wave function inside the core.

Field shift
Atomic-transition shift due to the change in nuclear charge radius, $F\,\delta\langle r^2\rangle$; here the relevant electron effect, and negligible.

6.3 Questions for discussion

1. The paper's headline is that the worst case is "unlikely". In Model A this is a statement about fine-tuning of unmeasured parameters. Can one attach a probability to it without a prior on $\beta_{3,4}$ and Δz? How would you choose that prior?
2. Model B reproduces ΔQ_0 to 7% but $\Delta\langle r^2\rangle$ only to a factor of 2. Is "renormalisation of the neutron magnetic moment in medium" a satisfactory explanation, and would it also change the prediction for ΔE_{SO} by a similar factor?
3. The two models are not obviously consistent with each other: Model A attributes ΔE_{em} to shape and skin differences of a charged drop, Model B to a spin–orbit term that Model A does not contain. Are they alternatives, or should they be added?
4. How robust is the proxy argument $K_s \sim \beta K_{\mathrm{em}}$? What would it take for ΔE_{em} and ΔE_{nuc} to scale identically with α_s? (The outlook admits this is open.)
5. The factor p in Eq. (16) is undetermined but called "order one". If p were 0.1, K would be $\sim 10^3$; still large. What experimental precision on neutron-capture cross sections would be needed to pin p to, say, 30%?
6. The 2024 $\Delta Q_0/Q_0$ value is close to the $\beta_{3,4} = 0$ funnel in Fig. 1 (left). Is it uncomfortable that the "simplest" Model A configuration sits near a zero?
7. From the experimental side: which of the requested measurements in Part 5 is most feasible at existing facilities, and which would you prioritise if the goal is to make K a number with an error bar?
8. Beyond α_{em} and α_s: what other parameters (quark masses, θ_{QCD}) could ULDM modulate, and does the K-enhancement carry over to them?

6.4 Suggested reading order for the original

1. Abstract and the last paragraph of the Introduction (the two questions).
2. "The ^{229}Th Nucleus" up to Eq. (4): make sure you can derive $K_{\mathrm{em}} \simeq \Delta E_{\mathrm{em}}/\Delta E$.
3. Fig. 1 and its caption *before* the geometric-model text; then read the text looking for the answers to Q1 and Q2.
4. The halo section: first the validation (Eqs. 11–13), then the energy (Eqs. 14–16).
5. Summary and Outlook; compare with the table in Part 5 of this guide.

Guide prepared from arXiv:2407.17526v2 (14 Sep 2025), the version matching the PRC Letter. The supplemental material (Fig. S1, spin–orbit moment and E_{SO} derivations) was not available and is not covered here.